

The primary goal of data warehousing is to access information that has high value of decision making. The data gathered from various sources.

The different ETL tool has been used to get data from heterogeneous sources and load into the target.

This article discusses the incremental and full refreshes using the different stored procedures. To increase the data freshness faster loading is essential.

Materialized view based on the base relations when base relation changes views needs to be changed. An incremental maintenance method for data warehouses allow improvements over the re-computation of views from scratch.

This article is more focused on the SQL performance optimization inside the stored procedures for incremental loading. In this stored procedure, the target table loads data from different source tables. For that, it also inserts timestamps along with data when the data gets inserted.

During the full refresh, the current data is excluded and it just inserts the new data from the staging table. Based on current time stamps, we fetch the next incremental load from the source table.

The existing method stores more resources including CPU and IO.

I have created a CustomerChurn table in SQL Server 2019.

USE [Data Warehouse]

GO

/***** Object: Table [dbo].[CustomerChurn_stage] Script Date: 9/4/2022

7:20:08 PM *****/

SET ANSI_NULLS ON

GO

```
SET QUOTED_IDENTIFIER ON
GO
```

```
CREATE TABLE [dbo].[CustomerChurn_stage](
    [CustomerId] [nvarchar](50) NOT NULL,
    [Surname] [nvarchar](50) NOT NULL,
    [CreditScore] [int] NOT NULL,
    [Geography] [nvarchar](50) NOT NULL,
    [Gender] [nvarchar](50) NOT NULL,
    [Age] [int] NOT NULL,
    [Balance] [float] NOT NULL,
    [Exited] [nvarchar](50) NOT NULL,
    CONSTRAINT [PK_CustomerChurn1] PRIMARY KEY CLUSTERED
(
    [CustomerId] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF,
IGNORE_DUP_KEY = OFF, ALLOW_ROW_LOCKS = ON,
ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF)
ON [PRIMARY]
) ON [PRIMARY]
GO
```

```
USE [Data Warehouse]
GO
```

/***** Object: Table [dbo].[CustomerChurn] Script Date: 9/4/2022 7:20:48

PM *****/

SET ANSI_NULLS ON

GO

SET QUOTED_IDENTIFIER ON

GO

CREATE TABLE [dbo].[CustomerChurn](

 [CustomerId] [varchar](50) NOT NULL,

 [Surname] [varchar](50) NULL,

 [CreditScore] [int] NULL,

 [Geography1] [nvarchar](50) NULL,

 [Gender] [nvarchar](20) NULL,

 [Age] [int] NULL,

 [balance] [float] NULL,

 [Exited] [nvarchar](50) NULL,

 [SourceSystemNM] [nvarchar](50) NOT NULL,

 [CreateAgentId] [nvarchar](20) NOT NULL,

 [CreateDtm] [datetime2](7) NOT NULL,

 [ChangeAgentId] [nvarchar](20) NOT NULL,

 [ChangeDtm] [datetime2](7) NOT NULL,

CONSTRAINT [PK_CustomerChurn_Stage] PRIMARY KEY CLUSTERED

(

 [CustomerId] ASC

```
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF,  
IGNORE_DUP_KEY = OFF, ALLOW_ROW_LOCKS = ON,  
ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF)  
ON [PRIMARY]  
) ON [PRIMARY]  
GO
```

```
alter PROCEDURE dbo.PrCustomerChurn  
as  
BEGIN
```

```
--*****  
*****
```

```
--DECLARE @VarCurrentTimestamp as  
TIMESTAMP=CURRENT_TIMESTAMP;--this built-in function might vary from  
database to database  
--DECLARE @VarCurrentTimestamp as  
TIMESTAMP=convert(timestamp,CURRENT_TIMESTAMP);  
DECLARE @VarCurrentTimestamp as datetime2=convert(datetime2,getdate());  
DECLARE @VarSourceRowCount as integer=0;  
declare @VarTargetRowCount as integer=0;  
declare @VarThresholdNbr as integer=0;
```

DECLARE @VarTinyIntVal as integer=0; --Note: TINYINT datatype can hold values between 0 and 255 only. Any value assignment outside this range would cause the stored procedure failure.

declare @tempdb as table(cnt int)

```
--*****  
*****
```

SELECT @VarSourceRowCount=COUNT(*)

FROM dbo.CustomerChurn_Stage; --Replace this with actual database name and table name (e.g., CustomerChurn_Stage) that you use.

select @VarSourceRowCount;

SELECT @VarTargetRowCount=count(*)

FROM dbo.CustomerChurn; --Replace this with actual database name and table name (e.g., CustomerChurn) that you use.

select @VarTargetRowCount

-- (TargetCount * 20%)

SELECT

@VarThresholdNbr=CAST(((@VarTargetRowCount * .2) AS INTEGER)

FROM @tempdb; -- The DUMMY is system table which might vary from database to database. For your database, you need to figure out.

select @VarThresholdNbr

```
--*****
```

--Fail the Stored Proc if the Source Row Count is less than the Threshold Number (i.e., 20% of the Target Table row count).

--This ensures that the Target table is not refreshed with incomplete set of Source Data

IF (@VarSourceRowCount < @VarThresholdNbr)

SELECT @VarTinyIntVal=-1

FROM @tempdb;

select @VarTinyIntVal

--*****

--*****

-- DELETE target table rows which are no longer available in source database
table.

DELETE

FROM dbo.CustomerChurn

WHERE EXISTS

(

SELECT *

FROM

(

SELECT dbo.CustomerChurn.CustomerId FROM dbo.CustomerChurn --AS

TrgtTbl -- Example table name: CustomerChurn

EXCEPT

```
SELECT dbo.CustomerChurn_Stage.CustomerId FROM
dbo.CustomerChurn_Stage --AS SrcTbl --Example table name:
CustomerChurn_Stage
```

```
) AS SrcTbl
```

```
WHERE CustomerChurn.CustomerId = SrcTbl.CustomerID)
--AND TrgtTbl.Primary Key Column2 = SrcTbl.Primary Key Column2
--AND ...
--*****UPDATE ROWS THAT CHANGED IN
SOURCE*****
-- Update the rows for which new version of rows have arrived as part of
delta/incremental feed (i.e., change to non-key values).
update
dbo.CustomerChurn
SET CustomerChurn.CustomerId = CustomerChurn_Stage.CustomerId,
    CustomerChurn.Surname= CustomerChurn_Stage.Surname,
    CustomerChurn.CreditScore = CustomerChurn_Stage.CreditScore,
    CustomerChurn.Geography1=CustomerChurn_Stage.Geography,
    CustomerChurn.Gender=CustomerChurn_Stage.Gender,
    CustomerChurn.Age=CustomerChurn_Stage.age,
    CustomerChurn.balance=CustomerChurn_Stage.balance,
    CustomerChurn.exited=CustomerChurn_Stage.exited,
    CustomerChurn.SourceSystemNM=
convert(nvarchar(50),SERVERPROPERTY('MachineName')),
--CustomerChurn.CreateDtm=@VarCurrentTimestamp,
    CustomerChurn.ChangeDtm = convert(datetime,GETDATE())
```

```
from dbo.CustomerChurn
```

```
INNER JOIN
```

```
dbo.CustomerChurn_Stage
```

```
on CustomerChurn.CustomerId = CustomerChurn_Stage.CustomerId
```

```
WHERE (
```

```
    COALESCE(CustomerChurn.Surname, '*') <>
```

```
COALESCE(CustomerChurn_stage.Surname, '*')
```

```
    OR COALESCE(CustomerChurn.CreditScore, '*') <>
```

```
COALESCE(CustomerChurn_stage.creditscore, '*')
```

```
    OR COALESCE(CustomerChurn.Geography1, '*') <>
```

```
COALESCE(CustomerChurn_Stage.Geography, '*')
```

```
    OR COALESCE(CustomerChurn.Gender, '*') <>
```

```
COALESCE(CustomerChurn_Stage.Gender, '*')
```

```
    OR COALESCE(CustomerChurn.Age, '*') <>
```

```
COALESCE(CustomerChurn_Stage.Age, '*')
```

```
    OR COALESCE(CustomerChurn.balance, '*') <>
```

```
COALESCE(CustomerChurn_Stage.balance, '*')
```

```
    OR COALESCE(CustomerChurn.Exited, '*') <>
```

```
COALESCE(CustomerChurn_Stage.Exited, '*')
```

```
)
```

```
-----create CustomerChurn table-----
```

```
--CREATE TABLE [dbo].[CustomerChurn](
```

```
--  [CustomerId] [varchar](50) NOT NULL,
```



```

-- [Surname] [varchar](50) NULL,
-- [CreditScore] [int] NULL,
-- [Geography1] [nvarchar](50) NULL,
-- [Gender] [nvarchar](20) NULL,
-- [Age] [int] NULL,
-- [balance] [float] NULL,
-- [Exited] [nvarchar](50) NULL,
-- [SourceSystemNM] [nvarchar](50) NOT NULL,
-- [CreateAgentId] [nvarchar](20) NOT NULL,
-- [CreateDtm] [timestamp] NOT NULL,
-- [ChangeAgentId] [nvarchar](20) NOT NULL,
-- [ChangeDtm] [datetime] NOT NULL,
-- CONSTRAINT [PK_CustomerChurn_Stage] PRIMARY KEY CLUSTERED
--(
-- [CustomerId] ASC
--)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF,
IGNORE_DUP_KEY = OFF, ALLOW_ROW_LOCKS = ON,
ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF)
ON [PRIMARY]
--) ON [PRIMARY]
insert into [CustomerChurn](
    [CustomerId],
    [Surname],
    [CreditScore],
    [Geography1],
    [Gender],
    [Age],

```

```

    [balance],
    [Exited],
    [SourceSystemNM],
    [CreateAgentId] ,
    [CreateDtm] ,
    [ChangeAgentId],
    [ChangeDtm])
SELECT
CustomerChurn_Stage.CustomerId
, CustomerChurn_Stage.Surname
---, SrcTbl.Primary Key Column3 ... list all PK columns based on your source/target
table primary key(s)
, CustomerChurn_Stage.CreditScore
, CustomerChurn_Stage.Geography
, CustomerChurn_Stage.Gender
, CustomerChurn_Stage.Age
, CustomerChurn_Stage.balance
, CustomerChurn_Stage.Exited
, convert(nvarchar(50), SERVERPROPERTY('MachineName')) AS
"SourceSystemNm" --Non-Primary Key
, USER AS "CreateAgentId" --Non-Primary Key -----use you own
USERID/USERNAME depending on the database system you use
, convert(datetime2, @VarCurrentTimestamp) AS "CreateDtm" --Non-Primary Key
, USER AS "ChangeAgentId" --Non-Primary Key -----use you own
USERID/USERNAME depending on the database system you use
, convert(datetime2, @VarCurrentTimestamp) AS "ChangeDtm" --Non-Primary
Key

```

FROM CustomerChurn_Stage

INNER JOIN

(

SELECT CustomerChurn_Stage.CustomerId

FROM CustomerChurn_Stage

EXCEPT

SELECT CustomerChurn.CustomerId

FROM CustomerChurn

) AS ChgdNew

ON CustomerChurn_Stage.CustomerId = ChgdNew.CustomerId

--AND SrcTbl.Primary Key Column2 = ChgdNew.Primary Key Column2

--AND ...

end;

print @VarSourceRowCount;

print @VarTargetRowCount;

print @VarThresholdNbr;

select * from tempdb;

Output of CustomerChurn table

100 %

Results Messages

	CustomerId	Surname	CreditScore	Geography1	Gender	Age	balance	Exited	SourceSystemNM	CreateAgentId	CreateDtm	ChangeAgentId	ChangeDtm
1	15568982	Hao	726	France	Female	24	0	0	LAPTOP-37L40S86	dbo	2022-09-04 19:10:46.7133333	dbo	2022-09-04 19:10:46.7133333
2	15569590	Yoo	601	Germany	Male	42	98495.71875	1	LAPTOP-37L40S86	dbo	2022-09-04 19:10:46.7133333	dbo	2022-09-04 19:10:46.7133333
3	15574012	Chu	645	Spain	Male	44	113755.78125	1	LAPTOP-37L40S86	dbo	2022-09-04 19:10:46.7133333	dbo	2022-09-04 19:10:46.7133333
4	15575185	Bushell	757	Spain	Male	33	77253.21875	0	LAPTOP-37L40S86	dbo	2022-09-04 19:10:46.7133333	dbo	2022-09-04 19:10:46.7133333
5	15577657	McDonald	732	France	Male	41	0	0	LAPTOP-37L40S86	dbo	2022-09-04 19:10:46.7133333	dbo	2022-09-04 19:10:46.7133333
6	15585768	Cameron	582	Germany	Male	41	70349.4765625	0	LAPTOP-37L40S86	dbo	2022-09-04 19:10:46.7133333	dbo	2022-09-04 19:10:46.7133333
7	15589475	Azikiwe	591	Spain	Female	39	0	1	LAPTOP-37L40S86	dbo	2022-09-04 19:10:46.7133333	dbo	2022-09-04 19:10:46.7133333
8	15592389	H?	684	France	Male	27	134603.875	0	LAPTOP-37L40S86	dbo	2022-09-04 19:10:46.7133333	dbo	2022-09-04 19:10:46.7133333
9	15592461	Jackson	603	Germany	Male	26	109166.3671875	0	LAPTOP-37L40S86	dbo	2022-09-04 19:10:46.7133333	dbo	2022-09-04 19:10:46.7133333
10	15592531	Bartlett	822	France	Male	50	0	0	LAPTOP-37L40S86	dbo	2022-09-04 19:10:46.7133333	dbo	2022-09-04 19:10:46.7133333

Output of CustomerChurn_Stage table

100 %

Results Messages

	CustomerId	Surname	CreditScore	Geography	Gender	Age	Balance	Exited
1	15568982	Hao	726	France	Female	24	0	0
2	15569590	Yoo	601	Germany	Male	42	98495.71875	1
3	15574012	Chu	645	Spain	Male	44	113755.78125	1
4	15575185	Bushell	757	Spain	Male	33	77253.21875	0
5	15577657	McDonald	732	France	Male	41	0	0
6	15585768	Cameron	582	Germany	Male	41	70349.4765625	0
7	15589475	Azikiwe	591	Spain	Female	39	0	1
8	15592389	H?	684	France	Male	27	134603.875	0
9	15592461	Jackson	603	Germany	Male	26	109166.3671875	0
10	15592531	Bartlett	822	France	Male	50	0	0

```

FROM [Data Warehouse].[dbo].[CustomerChurn]
order by customerid

select COUNT(*) from CustomerChurn_Stage;
select count(*) from CustomerChurn;

```

%

Results Messages

(No column name)
101

(No column name)
101

Fields which are present in backup table but not in the stage table

***** Script for SelectTopNRows command from SSMS *****/

SELECT TOP (1000) [CustomerId]

,[Surname]

,[CreditScore]

,[Geography]

,[Gender]

,[Age]

,[Balance]

,[Exited]

FROM [Data Warehouse].[dbo].[CustomerChurn_Backup]

except

```
SELECT [CustomerId]
      ,[Surname]
      ,[CreditScore]
      ,[Geography]
      ,[Gender]
      ,[Age]
      ,[Balance]
      ,[Exited]
FROM [dbo].[CustomerChurn_Stage]
```

100 %

Results Messages

	CustomerId	Surname	CreditScore	Geography	Gender	Age	Balance	Exited
1	15592389	H?	684	France	Male	27	134603.875	0
2	15604348	Allard	710	Spain	Male	22	0	0
3	15683553	O'Brien	788	France	Female	33	0	0
4	15687946	Osborne	556	France	Female	61	117419.3515625	0
5	15691483	Chin	549	France	Female	25	0	0
6	15701164	Onyeorulu	506	France	Female	34	90307.6171875	0
7	15725737	Mosman	669	France	Male	46	0	0
8	15737173	Andrews	497	Spain	Male	24	0	0
9	15755648	Pisano	675	France	Female	21	98373.2578125	0
10	15767821	Bearce	528	France	Male	31	102016.71875	0

Fields are present in Stage table but not in backup table

/***** Script for SelectTopNRows command from SSMS *****/

```
SELECT TOP (1000) [CustomerId]
      ,[Surname]
      ,[CreditScore]
      ,[Geography]
      ,[Gender]
```

,[Age]

,[Balance]

,[Exited]

FROM [Data Warehouse].[dbo].[CustomerChurn_Stage]

except

SELECT [CustomerId]

,[Surname]

,[CreditScore]

,[Geography]

,[Gender]

,[Age]

,[Balance]

,[Exited]

FROM [dbo].[CustomerChurn_backup]

Results Messages

	CustomerId	Surname	CreditScore	Geography	Gender	Age	Balance	Exited
1	15589975	Maclean	646	France	Female	73	97259.25	0
2	15592389	Vivek	684	India	Male	27	134603.875	0
3	15657566	Wieck	634	Germany	Male	24	103097.8515625	0
4	15683553	Osman	788	USA	Male	22	75888.296875	0
5	15691483	Chin	549	France	Female	25	12345.5	0
6	15698932	Groves	756	Germany	Male	44	137452.09375	0
7	15726676	Marshall	616	Spain	Male	30	0	0
8	15727556	O'Donnell	744	Spain	Female	26	166297.890625	0
9	15737173	Andrews	497	Spain	Male	29	0	1
10	15767821	Sharon	528	Hong Kong	Male	31	102016.71875	0
11	15771977	T'ao	730	France	Female	39	99010.671875	0